Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Hall, John L Ph.D.		
Project Title:	UltraStable Local Oscillator		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309-0440	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/15/2001	**End Date:**	02/14/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ye, Jun (JILA, University of Colorado)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

This task is pre-flight ground-based research to explore emerging new techniques in optical frequency measurement, atomic and molecular spectroscopy, and solid-state laser development. To advance space-based measurements using coherent optical sources, a continued improvement of laser frequency stability is needed. In a number of proposed space-based interferometric configurations, picometer resolution is the desired spatial quantity in maintaining a distance of millions of kilometers. This puts stringent requirements of the laser frequency stability on the order of 10-21. However, at a stability level of 10-15, powerful demonstrations of the utility of space interferometry-based measurements can already be realized, for example, in mapping the earth gravity field.
Our aim is to develop a laser system with an unprecedented stability, based on highly efficient new solid state lasers and improved precision molecular spectroscopy. We will refine current laser stabilization methods, leading to improved performance and a sharpened understanding of the current limitations and possibilities. The work will build on our

Task Description:	previous NASA funded research where we have demonstrated a stability of 5 x 10-15 at 300 s averaging time. Using new molecular transitions of narrower linewidths, we will study systematic effects such as recoil shift, power balance and signal modulation/extraction process. An optical frequency reference network will be established that offers wide tunability, while at the same time enjoys sub 10-15 ultrahigh stability. The system should prove to be extremely useful and adaptive to different tasks. Cross-Enterprise Project: Significance of Research Space Science Enterprise Space interferometry is the unifying theme of many programs under study or development within the Space Science Enterprise. To be able to collect light from scientific targets with a large aperture and then precisely combine different light beams interferometrically, it is necessary to have precision control of optical paths. Various levels of precision are needed, for example, Space Interferometry Mission (SIM) needs nanometer level control and stabilization of optical element positions to measure the relative positions and distances of stars: 1 nm is required to carry out successful experiments of star light nulling, 200 pm for wide-angle astrometry, and 50 pm for narrow-angle astrometry. Space Technology 3 (ST3) and Terrestrial Planet Finder (TPF) are missions to be carried out with separated spacecraft interferometry, leading to an elevated requirement on control and stability. Laser frequency stability on the order of 1 x 10-16 at 1 s will make these tasks realistic. The need for a more stable atomic clock will always be there as we try to the put the entire universe on a finer and finer map. Earth Science Enterprise Laser metrology between satellites will further improve the precision of enhanced formation flying. In the program of Gravity Recovery and Climate Experiment (GRACE), the Earth's gravity field will be mapped by measuring the along track separation between two spacecraft. A follow-on program, EX-5, calls for a drag-free spacecraft and distance tracking of below 100 nm. This enhanced capability will improve the sensitivity to changes in the Earth's gravity field by 2-3 orders of magnitude compared with GRACE. HEDS Enterprise The fundamental physics research program within High Energy Density Science (HEDS) can certainly benefit a great deal from the improved optical local oscillators. Many general relativity experiments such as the Laser Interferometer Space Antenna (LISA) and the Laser Interplanetary Ranging Experiment (LIRE) will need an unprecedented level of laser stability to carry out successful missions. Such achievement can be made possible only through the continued high level research of laser frequency stabilization. The proposed work should also make a strong impact to the cold atom research programs through the enhanced control of matter-field interaction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)